

Motasem ElGamel

Pittsburgh, PA

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Research Interests: Biophysics, Complex Systems, Stochastic Physics, Statistical Physics, Cell Size Control, and Microbial Ecology and Evolution.

Professional appointments

Tarbutton Postdoctoral Fellow

Theory and Modeling of Living Systems Initiative, Emory University
Advisors: Prof. Ilya Nemenman and Prof. Daniel Weissman.

2026 - Present

Atlanta, GA

Education

University of Pittsburgh

Physics - Ph.D.

Advisor: Prof. Andrew Mugler.

Dissertation: Microbial Dynamics at the Cellular and Population Levels.

2019 - 2025

Pittsburgh, PA

University of Pittsburgh

Physics - M.Sc.

2019 - 2021

Pittsburgh, PA

University of Science and Technology at Zewail City

Physics - B.Sc. (Graduated Magna Cum Laude)

2015 - 2019

Giza, Egypt

Publications

*Equal contribution

6. **M. ElGamel** and A. Mugler. Spatial structure promotes microbial resilience to environmental fluctuations. *In preparation.*
5. K. Ramachandran, **M. ElGamel**, and A. Mugler. Optimal information transmission in a sequential model for cell division. *In preparation.*
4. **M. ElGamel**, L. Ribaud, and A. Mugler. General mechanism for concentration-based cell size control. *arXiv preprint arXiv:2507.16066*, 2025.
3. S. Sayin*, **M. ElGamel***, B. Rosener, M. Brehm, A. Mugler, and A. Mitchell. Bacterial population dynamics during colonization of solid tumors. *Molecular Systems Biology*, 2025.
2. **M. ElGamel** and A. Mugler. Effects of molecular noise on cell size control. *Phys. Rev. Lett.*, 132:098403, 2024 - Press release ([link](#)).
1. **M. ElGamel**, H. Vashistha, H. Salman, and A. Mugler. Multigenerational memory in bacterial size control. *Physical Review E*, 108(3):L032401, 2023.

Workshops and Schools

NITMB and SMTPB Modeling and Theory in Population Biology Workshop

National Institute of Theory and Mathematics in Biology

- Competitively selected for a fully funded position.

June 2025

Chicago, IL

NITMB Ecological Dynamics of Microbial Communities: New Approaches Workshop

National Institute of Theory and Mathematics in Biology

May 2024

Chicago, IL

Les Houches Theoretical Biophysics Summer School 2023

École de physique des Houches

July 2023

Les Houches, France

- A month-long school aimed at elucidating the function and principles of biological systems, from cells to ecosystems, in light of the laws of physics.
- Worked on assessing selection and coevolution in protein sequences as part of a group project.

Talks and Presentations

1. Invited talks

Quantitative Biology Institute, Yale University

Invited talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria"

May 2025

New Haven, CT

Department of Physics, Emory University

Invited talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria"

April 2025

Atlanta, GA

The National Institute for Theory and Mathematics in Biology

Invited talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria"

December 2024

Chicago, IL

Group meeting of Prof. Michael Stumpf (virtual)

Invited talk: "Effects of molecular noise on cell size control"

November 2024

The University of Melbourne, Australia

SIAM Conference on the Life Sciences 2024

Invited talk: "Clone size statistics of tumor-inhabiting bacteria"

June 2024

Portland, OR

2. Selected talks

NITMB and SMTPB Modeling and Theory in Population Biology Workshop

Selected talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria".

June 2025

Chicago, IL

Gordon Research Conference and Seminar

Selected talk and poster presentation: "Population dynamics and universal statistics of tumor-inhabiting bacteria".

January 2025

Ventura, CA

APS March Meeting 2024

Selected talk for DBIO Early Career Prize Session: "Clone size statistics of tumor-inhabiting bacteria".

March 2024

Minneapolis, MN

Gordon Research Conference and Seminar

Selected talk and poster presentation: "Effects of molecular noise on cell size control".

January 2023

Ventura, CA

NSF Center for the Physics of Biological Function - Physics of Life Symposium

Selected talk: "Theory of cell size homeostasis".

November 2022

New York City, NY

The 2nd Biology for Physics Conference

Selected talk: "Multigenerational memory in bacterial size control".

July 2022

Barcelona, Spain

3. Contributed, seminar, and poster presentations

Pitt Biological Physics Day

Poster presentation: "General mechanism for concentration-based cell size control".

September 2025

Pittsburgh, PA

APS Global Physics Summit 2025

Contributed talk: "Population dynamics and universal statistics of tumor-inhabiting bacteria".

March 2025

Anaheim, CA

Pitt Biological Physics Day

Poster presentation: "Population dynamics and universal statistics of tumor-inhabiting bacteria".

September 2024

Pittsburgh, PA

NITMB Ecological Dynamics of Microbial Communities: New Approaches Workshop

Lightning talk: "Clone size statistics of tumor-inhabiting bacteria"

May 2024

Chicago, IL

University of Pittsburgh Grad Expo 2024

Contributed talk: "Clone size statistics of tumor-inhabiting bacteria"

March 2024

Pittsburgh, PA

Applied Math Seminar, Department of Mathematics, University of Pittsburgh

Seminar: "Growth Dynamics of Tumor-Inhabiting Bacteria"

March 2024

Pittsburgh, PA

Les Houches Theoretical Biophysics Summer School 2023

Poster presentation: "Theory of noisy population growth in bacteria".

July 2023

Les Houches, France

Physics of Life 2023 Conference

Poster presentation: "Effects of molecular noise on cell size control".

March 2023

Harrogate, UK

APS March Meeting 2023

Contributed talk: “Adder minimizes cell size noise in bacteria”.

March 2023

Las Vegas, NV

APS March Meeting 2022

Contributed talk: “Multigenerational memory in cell size homeostasis”.

March 2022

Chicago, IL

Fellowships and Awards

Tarbutton Postdoctoral Fellowship

2026 -

Theory and Modeling of Living Systems Initiative, Emory University

APS FIP Distinguished Student Travel Award

2026

American Physical Society (APS) and Forum on International Physics (FIP)

Selected Speaker at the DBIO Early Career Prize Session

2024

American Physical Society (APS) March Meeting 2024

Outstanding Presentation Award Pitt Grad Expo 2024

2024

Dietrich School of Arts and Sciences, University of Pittsburgh

- Awarded to the talk: “Clone size statistics of tumor-inhabiting bacteria”.

Andrew W. Mellon Predoctoral Fellowship

2023 - 2024

University of Pittsburgh

- Awarded school-wide to doctoral students of exceptional promise and ability across the disciplines based on a research proposal. Consists of funding for one year.

Best Poster Award at the Physics of Life 2023 Conference, Harrogate, UK

2023

- Awarded to the poster presentation: “Effects of molecular noise on cell size control”.

Pitt Graduate and Professional Student Government (GPSG) Travel Grant

2023

University of Pittsburgh

Thomas-Lain Fund Scholarship Essay Competition

2023

Department of Physics and Astronomy, University of Pittsburgh

Mentorship

Research Mentor

Department of Physics and Astronomy, University of Pittsburgh

- Mentored undergraduate students Jianan Zhao and Lucas Ribaud* in research.

* Awarded the Emil Sanielevici Undergraduate Research Scholarship for research on concentration-based cell size control in yeast.

Peer-Mentor in ZC Alumni Mentorship Program

Zewail City Alumni Association

- Mentored one undergraduate student.

Graduate Student Mentor

Department of Physics and Astronomy, University of Pittsburgh

- Mentored two incoming graduate students.

Teaching

Graduate Teaching Assistant

University of Pittsburgh

- PHYS 0110: Introduction to Physics 1 - PHYS 0111: Introduction to Physics 2 - PHYS 0174: Basic Physics for Science and Engineering 1.
- Lead recitations, held office hours, prepared quizzes, graded exams and wrote up homework solutions.

Undergraduate Junior Teaching Assistant

University of Science and Technology at Zewail City

- PHYS 101: Introduction to Physics 1 - PEU 438: Compact Objects.

Outreach and Service

Journal Reviewer

- Reviewed articles for *Physical Review Letters*, *Physical Review E*, *Physical Review X Life* (with advisor), *Physical Review X* (with advisor), and *Springer's Bulletin of Mathematical Biology*.

Co-chair of the Stochastic Physics in Biology Gordon Research Seminar

January 2027

- Organizing the next Stochastic Physics in Biology Gordon Research Seminar in 2027.

Ventura, CA

Lecturer at the 1st Zewail City Alumni Physics Winter School (virtual)

January 2025

University of Science and Technology at Zewail City

Zewail City, Egypt

- Lecture: "How Do Cells Navigate a Noisy Life?".
- General introductory lecture on biological physics.

Zewail City Science Festival

2016 - 2017

University of Science and Technology at Zewail City

Zewail City, Egypt

- Event that aims to deliver scientific concepts in a simplified manner to the public and spread the culture of science.
- Participated as an organizer, wrote a scientific talk for the public audience, and gave a mathematics talk.

Skills

Programming Languages: C++, Python, Matlab, Mathematica, LabVIEW.

Certificates: Neural Networks and Deep Learning ([link](#)).